TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Operation of Sweater Knitting Machines and Visit to a Sweater Manufacturing Plant

Hands on the V-bed flat machine — setting up a fashioned panel programme and reading its shaping — followed by a structured plant visit that traces one order through knitting, linking, washing, boarding and packing.

FM-4202 · ADVANCED KNITTING SESSIONAL · EXPERIMENT 4 · FABRIC MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to operate the laboratory flat knitting machine through the production of a fashioned panel, and to observe and record the complete sweater factory process during a plant visit.

The V-bed machine's theory is FM-4201 Topic 2's: two inclined beds, latch needles, a traversing carriage whose cams knit, tuck, miss and transfer; shaping by transfer at panel edges (fashioning). Operation adds the practical layer the theory omits: the programme is a list of instructions (courses, yarn carriers, stitch values, shaping events), the panel is checked by **weight and dimensions against the spec while it is still on the machine's side of the factory**, and every later stage (linking, washing) is designed around what knitting delivers. The plant visit then reads the factory as one material flow with quality gates — the same discipline as a weaving-shed study, applied to knitwear.

2. Apparatus

- Laboratory V-bed flat knitting machine with programming station; yarn of the panel spec; panel spec sheet (measurement chart); balance and measuring tape; takedown/pressing access.
- For the visit: factory permission and safety briefing, observation checklist (Section 4), notebook — photographs only where the factory permits.

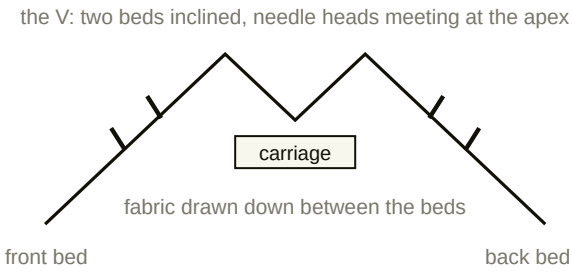


Figure 1 — V-bed geometry in section (schematic): needles from the two beds cross at the apex to knit and transfer; the carriage traverses above, the panel descends between the beds to the takedown.

3. Procedure — machine operation

1. **Programme study:** load the panel programme; read it line by line against the spec: cast-on, rib border courses, transfer to main bed, body courses with stitch values, shaping events (position, frequency, times), bind-off. Predict the panel's stitch count at three heights before knitting.
2. **Setup:** mount and thread the yarn carriers per programme; check needle condition in the working width (Experiment 2's eye, flat-bed edition); set takedown and stitch values to the programme.
3. **Knit and witness:** run the panel; observe transfer at a shaping event (needles aligning across the V, stitches handed over); stop at any fault and record it as the factory would (cause, courses lost).
4. **Check the panel:** weigh it (against the spec weight), count courses, measure width/length off the machine; compare with the predictions of step 1 and with the finished-garment dimensions the panel must reach after washing (Experiment 5's shrinkage allowances).

WORKED CALCULATION — PANEL STITCH COUNT FROM THE SPEC

Spec: front panel finished width 50 cm, machine 7 gg.
Stitches across = 50 × (7 ÷ 2.54) = 50 × 2.76 ≈ **138 stitches** — the working width the programme must cast on (before selvedge/shaping allowances), and the number checked against the loaded programme in step 1.
The same conversion, run at the plant's gauges during the visit, lets every measurement chart be read as a machine instruction — the visit's core skill.

4. Procedure — the plant visit (observation checklist)

1. **Yarn store and preparation:** lot control, conditioning, winding/rewinding, yarn testing records. Note how lots are kept apart (shade continuity between panels of one garment).
2. **Knitting section:** machine park (gauges, ages, electronic vs mechanical), programming office and its file discipline, panel weight checking at the machine, fault recording and efficiency display.

3. **Linking and make-up:** linking machines and operator skill system, seaming types used, trim application; observe how fashioned marks are matched at seams.
4. **Washing and finishing:** wash types (relaxation only vs milling), drying, boarding/pressing lines, measurement control point and its sampling plan.
5. **Quality and packing:** final inspection method, AQL use (compare TE-3107), shade-band control, folding/packing and the records that travel with the order.
6. **Write-up:** process flow of the plant as one diagram, with each quality gate marked, throughput notes (orders in work, lead stages), and three observed strengths and three observed risks, each tied to a mechanism from this course — not to impressions.

5. Observation table (form — model entries)

STAGE	OBSERVATION (MODEL)	QUALITY GATE
Knitting	Mixed park: electronic V-bed for fashioning, mechanical for basics; panel weight checked 1 in 5 panels	Panel weight + course count at machine
Linking	Loop-to-loop linking on shaped seams; overlock on straight joins	Seam appearance sampling by line chief
Washing	Relaxation wash standard; milling only on specified wool orders	Post-wash measurement vs chart
Boarding	Form boarding for sweaters; steam press for cardigans	Measurement table audit per bundle
Final	100% visual + sampled measurement; AQL for shipment release	AQL sampling record

6. Precautions

- Flat machine: hands clear of the traversing carriage path; stop before reaching for any fault — the carriage returns without asking.
- During the visit: follow the host's safety route, no touching of machines or panels in production, no photographs beyond permission.
- Record only what was observed; where the host's account and the floor disagree, note both.

7. Practice (viva preparation)

1. Why is panel weight, rather than panel length, the knitter's first check?
2. Convert a 55 cm chest-panel width to stitches on a 12 gg machine.
3. What does the linking operator use the fashioning marks for?
4. Name the stage where a sweater's final dimensions are actually fixed, and the knitting consequence.
5. Give two plant-visit observations that would count as *risks*, each with its mechanism.

Answer key

1. Weight integrates yarn count, stitch length and total courses knitted in one instant measurement: a panel can be the right length and the wrong everything else (wrong stitch value compensated by wrong course count), but it cannot be the right weight by accident.
2. 55 × (12 ÷ 2.54) = 55 × 4.72 ≈ **260 stitches**.
3. Alignment: the marks are identical, counted features on both panels — matching mark to mark guarantees the two sides are joined loop-for-loop in register, so the seam neither twists nor creeps.
4. **Washing (with boarding)**: relaxation and milling set the final dimensions; consequently knitting targets are deliberately written oversize/undersize by the known shrinkage allowances — a knitting spec is a pre-wash document (Experiment 5 measures exactly those allowances).
5. Examples: (i) *Yarn lots mixed at the creel* — mechanism: panels of one garment from different dye lots will band in shade after washing; the fault is created at knitting and only visible at final inspection. (ii) *Panel weight unchecked on the night shift* — mechanism: stitch-value drift accumulates into systematic dimension failure discovered only after linking, when rework is dearest. (Any two with a mechanism earn the marks.)
